Year 13 Further Maths - Further calculus techniques

Similar questions - Improper integrals

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning. Determine the exact value of $\int_{16}^{\infty} \frac{12}{x^2\sqrt{x}}\,\mathrm{d}x$. **[4]**

2

(a) Show that $\int_{2}^{\infty} (x-1)^{-2}\,\mathrm{d}x = 1$. **[3]**

(b) Show that $\int_{1}^{\infty} \frac{1}{\sqrt{x}}\,\mathrm{d}x$ does not converge. **[2]**

(c) Evaluate $\int_{0}^{1} \frac{1}{\sqrt{x}}\,\mathrm{d}x$, explaining why it is an improper integral. **[3]**

Question 1

$\int_{16}^{k} 12x^{-5/2}\,\mathrm{d}x = \left[-8x^{-3/2}\right]_{16}^{k} = -8k^{-3/2} + \frac{8}{64}$. As $k \to \infty$, $k^{-3/2} \to 0$, so the value is $\frac{1}{8}$

Question 2

(a) $\int_2^k (x-1)^{-2}\mathrm{d}x = \left[-(x-1)^{-1}\right]_2^k = 1 - \frac{1}{k-1} \to 1$ as $k \to \infty$

(b) $\int_1^k x^{-1/2}\mathrm{d}x = 2\sqrt{k} - 2 \to \infty$ as $k \to \infty$, so the integral diverges

(c) The integrand is undefined at $x = 0$. $\int_h^1 x^{-1/2}\mathrm{d}x = 2 - 2\sqrt{h} \to 2$ as $h \to 0^+$, so the value is 2